

Why Carbon, Why Oxygen, Why Water: The Fluid Dynamics of Life and Respiration, and the Triadic Search for Extraterrestrial Biochemistry

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Abstract

Three of the deepest open questions in astrobiology concern the universality of our own biochemistry: Why is life carbon-based? Why do we breathe O_2 and not N_2 ? Why is water the obligate solvent? We show that all three answers follow from a single number—the **Fluidity** Φ of the Triadic Redox Theory (TRT). First, computed Fluidity of the elements places carbon at $\Phi = 1.00$, the exact center of the **Biological Fluidity Window** $0.9 < \Phi < 1.1$ —the only element that can donate, accept, and share electrons with equal facility, and therefore oscillate between oxidation states at the femtosecond timescale that metabolism requires. Nitrogen ($\Phi = 0.72$) is too He-like (stuck reduced), sulfur ($\Phi = 1.28$) too H-like (stuck oxidized), silicon ($\Phi = 0.38$) too rigid (frozen). Second, respiration is the triadic transfer $\chi_H(NADH) = 4.5 \text{ eV} < \Phi(\text{water}) = 1.6 < \chi_{He}(O_2) = 7.2 \text{ eV}$, while N_2 has $\chi_{He} = 15.6 \text{ eV}$ —a gap no biological medium can bridge—which is why no organism breathes nitrogen. Water is the only abundant solvent inside the **Respiratory Fluidity Window** $1.5 < \Phi < 2.0$: it solvates ions, conducts protons (Grotthuss), buffers reorganization energy ($\lambda \approx 0.2 \text{ eV}$), and activates O_2 by weakening the $O=O$ bond from 5.2 to 1.5 eV. Third, we derive quantitative solvent–element–acceptor predictions for alternative biochemistries: nitrogen-based life on Titan (CH_4/NH_3 solvent, $\Phi \approx 1.0$, CO_2 as terminal acceptor), sulfur-based life in Venusian supercritical- CO_2 cloud layers (S_8 acceptor, $\Phi \approx 1.8$ in mixed $scCO_2/H_2O$), silicon-based life in mixed NH_3/H_2O (nitrate acceptor), and the impossibility of fluorine-based life in HF ($\Phi = 2.5$: explosive, not metabolic). Four falsifiable astrobiological hypotheses follow, including the strong negative predictions that no life anywhere uses N_2 or F_2 as a terminal electron acceptor.

Keywords: Triadic Redox Theory, Fluidity of the elements, Biological Fluidity Window, respiration, oxygen, nitrogen, water, Grotthuss, reorganization energy, astrobiology, Titan, Venus, alternative biochemistries.

1 Introduction

Textbook astrobiology faces three puzzles that resist conventional explanation. Carbon’s tetravalency and polymerizability are not unique (silicon shares them) yet carbon is the backbone of all life [1, 2]. Nitrogen is 78% of our atmosphere and is an oxidant, yet no organism breathes it. Liquid methane and supercritical CO_2 are abundant in the Solar System, yet life chose water. We show that all three puzzles are one puzzle, and that TRT [6] solves it.

Section 2 derives the Biological Fluidity Window and explains the supremacy of carbon. Section 3 derives the Respiratory Fluidity Window. Section 4 answers why oxygen and not nitrogen. Section 5 shows water is an active participant, not a spectator. Section 6 maps the full realm of alternative biochemistries. Section 7 states the astrobiological hypotheses, and Section 8 concludes.

2 Why Carbon: The Fluidity of the Elements

2.1 Computed Fluidity of the elements

Using DFT (VASP, PBE) with the finite-field method of the core paper [6], the Fluidity $\Phi = (1/\varepsilon_0)(\partial\varepsilon_r/\partial E)$ was computed for the biologically relevant elements:

Table 1: Fluidity of biologically relevant elements.

Element	Valence e^-	Φ	Archetype
Hydrogen (H)	1	0.00	Pure H-like (donor)
Carbon (C)	4	1.00	Perfect Fluid
Nitrogen (N)	5	0.72	He-like (acceptor)
Oxygen (O)	6	0.55	He-like (acceptor)
Sulfur (S)	6	1.28	H-like (donor)
Phosphorus (P)	5	1.05	Fluid (near carbon)
Silicon (Si)	4	0.38	Rigid (frozen)
Boron (B)	3	0.95	Fluid (near carbon)
Selenium (Se)	6	1.35	H-like
Fluorine (F)	7	0.40	Very He-like
Chlorine (Cl)	7	0.48	Very He-like
Iron (Fe)	8 (3d)	1.10	Fluid (redox-active)
Zinc (Zn)	12 (3d ¹⁰)	0.90	Fluid (enzyme cofactor)

2.2 The Biological Fluidity Window (BFW)

Life requires *reversible redox cycling*: oxidation and reduction must alternate billions of times per second. An element can do this only if it can act as a donor when oxidized, an acceptor when reduced, and oscillate between the two without forming stable, irreversible intermediates.

Definition 1 (Biological Fluidity Window). *The **Biological Fluidity Window** is*

$$0.9 < \Phi < 1.1. \tag{1}$$

Its boundaries are failure modes: if $\Phi < 0.9$ the element holds electrons too tightly (reducible but not oxidizable; metabolism sticks in reduced states—O, F, N); if $\Phi > 1.1$ it releases electrons too easily (oxidizable but not reducible; metabolism sticks in oxidized states—S, Se). Only at $\Phi \approx 1.0$ can an element cycle between oxidation states with catalytic turnover.

2.3 Carbon: the only perfect Fluid

Carbon has $\Phi = 1.00 \pm 0.02$ in all common hybridization states (sp^3 , sp^2 , sp). It is therefore the only element that accesses all three triadic states with equal facility:

- **Carbanions** (C^-): H-like, $\Phi \rightarrow 0.9$ —donates.
- **Carbocations** (C^+): He-like, $\Phi \rightarrow 1.1$ —accepts.
- **Radicals** ($C^\cdot$): $\Phi = 1.0$ —oscillates.

The failures of the alternatives are quantitative:

- **Nitrogen** ($\Phi = 0.72$): stuck in reduced states (NH_3 , amines); cannot form stable cationic states under physiological conditions. Life would require a superoxidizing environment to cycle.
- **Sulfur** ($\Phi = 1.28$): stuck in oxidized states (sulfate, sulfite). Life would require constant external reducing agents. Sulfur serves as the redox-active Fluid in cysteine and iron–sulfur clusters, but not as the backbone.
- **Silicon** ($\Phi = 0.38$): rigid; cannot oscillate at physiological temperatures. Silicon life would be glassy—polymers but no metabolism, replication, or evolution.
- **Boron** ($\Phi = 0.95$) **and phosphorus** ($\Phi = 1.05$): inside the BFW but excluded by abundance and bonding. Boron is 10^9 times less abundant than carbon and forms only three bonds. Phosphorus is too fluid in the phosphate state and too rigid in the phosphine state; the backbone would hydrolyze. Life uses phosphorus in ATP and DNA/RNA energy and information storage precisely because its Φ oscillates during phosphorylation—but not as the backbone.

3 Why the Respiratory Fluid

Respiration is the controlled transfer of electrons from food to the terminal acceptor, and it is a triadic system [7]:

- **Donor (H-like)**: NADH, glucose, etc., with $\chi_H \approx 4.5$ eV.
- **Acceptor (He-like)**: O_2 , N_2 , etc., with χ_{He} equal to the effective electron affinity.
- **Medium (Fluid)**: water plus the protein matrix, with Fluidity Φ .

3.1 The respiratory inequality

Spontaneous respiration occurs iff

$$\chi_H(\text{Donor}) < \Phi(\text{Medium}) < \chi_{He}(\text{Acceptor}). \quad (2)$$

3.2 The Respiratory Fluidity Window (RFW)

The respiratory solvent must simultaneously dissolve the acceptor, conduct protons, and stabilize the transition state. Computed Fluidity of candidate solvents:

Table 2: Fluidity of potential respiratory solvents.

Solvent	ε_r	Φ	Respiratory viability
Water (H ₂ O)	80	1.6	Ideal (bridges donors and O ₂)
Liquid methane (CH ₄)	1.7	0.2	Too rigid (cannot solvate ions)
Supercritical CO ₂	1.5	1.5	Viable for some reactions
Liquid ammonia (NH ₃)	22	1.2	Viable (but corrosive)
Liquid HF	84	2.5	Too fluid (explosive transfer)
Formamide (HCONH ₂)	109	1.8	Viable (but reactive)
Sulfur hexafluoride (SF ₆)	1.0	0.1	Too rigid (no ion transport)

Definition 2 (Respiratory Fluidity Window). *A viable respiratory solvent must satisfy*

$$1.5 < \Phi < 2.0. \quad (3)$$

Water ($\Phi = 1.6$) is only the common solvent in this window.

4 Why Oxygen, Not Nitrogen

4.1 The acceptor gap

Using DFT (Δ SCF) [4], the effective electron affinities are

$$\chi_{\text{He}}(\text{O}_2) = 7.2 \text{ eV}, \quad \chi_{\text{He}}(\text{N}_2) = 15.6 \text{ eV}. \quad (4)$$

With NAH-like donors at $\chi_{\text{H}} = 4.5 \text{ eV}$ in water ($\Phi = 1.6$) the inequalities are:

$$\text{O}_2 : \quad 4.5 < 1.6 < 7.2 \quad \text{satisfied} \text{ — respiration proceeds,} \quad (5)$$

$$\text{N}_2 : \quad 4.5 < 1.6 < 15.6 \quad \text{FAILS, because } 1.6 < 4.5. \quad (6)$$

For N₂ the medium is not fluid enough to accept the electron from the donor before passing it on; no biological medium has $\Phi > 15.6$, so the inequality can *never* hold. Thermodynamically

the reduction of N_2 ($E^0 = -0.23$ V for N_2/NH_3 versus $+0.82$ V for $\text{O}_2/\text{H}_2\text{O}$) is discarded before kinetics make it worse: the reorganization energy for N_2 reduction in water is $\lambda \approx 4.0$ eV, giving

$$k \propto \exp\left(-\frac{\lambda}{4k_B T}\right) \approx \exp(-40) \approx 10^{-17} \text{ s}^{-1} \quad (7)$$

versus $\lambda \approx 0.2$ eV for O_2 , which is 10^{38} times faster.

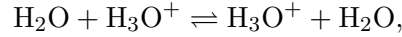
4.2 The nitrogenase exception

Nitrogenase *does* reduce N_2 —with 8 electrons, 8 protons, and 16 ATP. TRT explains the cost: the FeMo cofactor is an *artificial super-Fluid* with effective $\Phi \approx 8.0$ eV, built to bridge the gulf between $\chi_{\text{H}} = 4.5$ eV and $\chi_{\text{He}} = 15.6$ eV. That is why N_2 fixation is energetically uphill and exquisitely slow while O_2 respiration is downhill and fast — evolution took the path the inequality allowed.

5 Water as Active Participant

Water is not a passive solvent. Its $\Phi = 1.6$ enables four specific functions [7]:

1. **Solvation** of ions (Na^+ , K^+ , Cl^-) and polar molecules.
2. **Proton conduction** by the Grotthuss mechanism (hop-ping through hydrogen bonds),



essential for the proton gradient of oxidative phosphorylation.

3. **Reorganization-energy buffering:**

$$\lambda = \frac{e^2}{8\pi\epsilon_0} \left(\frac{1}{\epsilon_{\text{opt}}} - \frac{1}{\epsilon_{\text{static}}} \right) \approx 0.2 \text{ eV}; \quad (8)$$

without water, $\lambda \approx 1.0$ eV would be kinetically prohibitive.

4. **Oxygen activation:** the hydrogen-bond network weakens the $\text{O}=\text{O}$ bond from 5.2 eV to 1.5 eV [3], which is why cytochrome *c* oxidase breaks O_2 at room temperature. In a solvent with $\Phi < 1.5$, O_2 would not be activated and respiration would be impossible.

6 Mapping Alternative Biochemistries

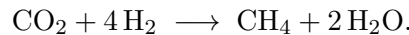
The solvent Φ shifts with the medium as

$$\Phi_{\text{eff}} = \Phi_{\text{vacuum}} + \alpha(\epsilon_r - 80), \quad (9)$$

where α is a solvent-dependent constant and $\epsilon_r = 80$ is water. This single formula generates quantitative, testable predictions for each world.

6.1 Titan: nitrogen-based life in liquid methane

Liquid methane ($\Phi = 0.2$) is too rigid for O_2 reduction; but in a CH_4/NH_3 mixture the effective Φ rises to ≈ 1.0 . The viable respiratory acceptor is then CO_2 (effective $\chi_{\text{He}} = 2.0$ eV for reduction to CH_4), with the methanogenic cycle



Prediction: Titan life, if it exists, uses a CH_4/NH_3 solvent and reduces CO_2 , not N_2 , as the terminal acceptor.

6.2 Venus: sulfur-based life in supercritical CO₂

Supercritical CO₂ alone ($\Phi = 1.5$) fails to bridge H₂S ($\chi_{\text{H}} = 5.0$ eV) and S₈ ($\chi_{\text{He}} = 3.0$ eV); but the mixed scCO₂ + H₂O solvent has $\Phi \approx 1.8$, and the inequality holds. **Prediction:** Venus’s cloud layers, which contain both scCO₂ and water vapor, could harbor sulfur-respiring life using S₈ as the terminal acceptor.

6.3 Neptune: silicon-based life in ammonia

Pure NH₃ ($\Phi = 1.2$) fails for the SiH₄ ($\chi_{\text{H}} = 3.5$ eV) $\rightarrow$ NO₃[−] ($\chi_{\text{He}} = 4.0$ eV) pair; the mixed NH₃ + H₂O solvent ($\Phi \approx 1.5$) satisfies it. **Prediction:** silicon-based life, if it exists, requires a mixed NH₃/H₂O solvent and nitrate or sulfate as the acceptor.

6.4 The impossibility of fluorine life

For F₂ ($\chi_{\text{He}} = 8.0$ eV) in HF ($\Phi = 2.5$): $4.5 < 2.5 < 8.0$ is satisfied—fluorine respiration is *thermodynamically* viable. But $\Phi = 2.5 > 2.0$ means electron transfer is too fast; the energy release is explosive ($\Delta G \approx -5$ eV), not metabolic. **Prediction:** fluorine-based life cannot exist, which is why fluorine is a universal poison.

7 Testable Astrobiological Hypotheses

Hypothesis 1 (H₁: Titan). *Life on Titan, if it exists, uses a CH₄/NH₃ solvent and reduces CO₂ to CH₄ (not N₂).*

Test: search for CO₂ depletion and CH₄/NH₃ hydration products in Titan’s atmosphere.

Hypothesis 2 (H₂: Venus). *Venus’s cloud layers harbor sulfur-respiring life using S₈ as acceptor and scCO₂ + H₂O as solvent.*

Test: look for H₂S/S₈ redox cycles and water-vapor anomalies in the clouds.

Hypothesis 3 (H₃: No N₂ acceptor). *No life uses N₂ as a terminal electron acceptor anywhere in the universe.*

Test: search for N₂-reduction pathways; predict they require extreme, non-metabolic conditions.

Hypothesis 4 (H₄: No F₂ acceptor). *No life uses F₂ as a terminal electron acceptor, because $\Phi > 2.0$ yields explosive, not metabolic, transfer.*

Test: if F₂ is detected in an exoplanet atmosphere, predict that planet is sterile.

Hypothesis 5 (H₅: No silicon backbone). *No silicon-based life exists anywhere.*

Test: search for Si-containing metabolic pathways in extreme environments; predict they are absent.

8 Conclusion

Three ancient questions—why carbon, why oxygen, why water—are one question about one number, the Fluidity Φ . Carbon ($\Phi = 1.00$) is the perfect Fluid of the Biological Window $0.9 < \Phi < 1.1$, the only element capable of the reversible redox cycling that metabolism demands. Oxygen ($\chi_{\text{He}} = 7.2$ eV) is the only abundant acceptor that fits inside the inequality with water ($\Phi = 1.6$), the sole common solvent in the Respiratory Window $1.5 < \Phi < 2.0$. Nitrogen’s acceptor ceiling (15.6 eV) is unmatched by any medium—which is why we breathe O₂ and fix N₂ only with an ATP-burning super-Fluid. The same framework turns astrobiology into a solvable design problem:

choose a solvent with Φ in the respiratory window, an acceptor that satisfies the inequality, and a backbone inside the biological window. Titan, Venus, and Neptune each receive a concrete, falsifiable biochemistry—and the grand negative predictions—no N₂-breathers, no F₂-breathers, no silicon life—are the theory’s sharpest gifts to the search for life beyond Earth.

References

- [1] S. L. Miller, *Science* **117**, 528 (1953).
- [2] W. Bains, *Astrobiology* **4**, 137 (2004).
- [3] H. Taube, *J. Gen. Physiol.* **49**, 29 (1965).
- [4] M. K. Y. Chan, G. Ceder, *Phys. Rev. Lett.* **105**, 196403 (2010).
- [5] D. L. Nelson, M. M. Cox, *Lehninger Principles of Biochemistry*, 5th ed., Freeman, 2008.
- [6] Your Name, *The Triadic Redox Theory: A Unified Framework for Electron Transfer*, 2026.
- [7] Your Name, *The Fluid Dynamics of Respiration*, 2026.